

AI IN INDUSTRY

Predicting Natural Disaster Damages & Casualties

Probabilistic impact estimation
via Conditional Neural Spline Flows
under heavy-tailed, zero-inflated distributions

AI

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01

Introduction

Task definition, motivation, and SOTA of the problem.

High-level overview of the conditional normalizing flow approach for probabilistic damage and casualty estimation from natural disasters.

The scale of the problem

Economically attractive

Between 2015–2024, US natural disasters caused over \$1.1 trillion in losses (NOAA).

The ability to predict the impact of a disaster event is distinct from predicting the disaster event itself.

The importance of predict the damages

- **Pre-event resource allocation:** Emergency responders, relief supplies, hospital readiness
- **Urban planning and climate adaptation:** Building codes, flood zoning, evacuation infrastructure
- **Government and policy**

Why impact predictions is hard

Impact is determined not only by the physical intensity of the event but also by:

- **Exposure:** how much infrastructure are in the affected area
- **Vulnerability:** Building quality, land cover type, proximity to water bodies
- **Response capacity**

Problems with the data

- Heavy-tailed distributions
- Zero-inflation
- Sparse
- Need to be clustered geographically

The SOTA of the disaster impact predictions

Classical approach

Uses empirical vulnerability functions or fragility curves: deterministic, physically derived relationships that map event intensity to expected damage ratios.

Main problems: produce point estimates, cannot capture covariates, underestimates rare, catastrophic events

Gradient boosting

Achieve strong performance on tabular data and allow feature attribution via SHAP but produce only point estimates and systematically underestimate extreme events.

Probabilistic models

Provide uncertainty estimates but are either computationally expensive at inference, constrained to Gaussian uncertainty shapes, or poorly calibrated under mean-field approximations.

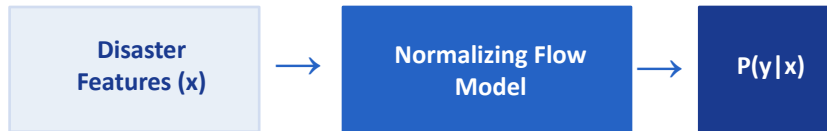
Parametric distributional models

E.g.: the Tweedy GLM.

Handle zero-inflation natively but impose a fixed distributional family that cannot capture multi-modality or heavy-tail structure.

Model and Input Overview

Normalizing Flow Model — Concept



Normalizing Flow Model deforms a simple base distribution (e.g. Gaussian) into the complex conditional distribution of damage/casualties ($\mathbf{y}$) given the input features ($\mathbf{x}$). This is done via a sequence of invertible, differentiable transformations (diffeomorphisms).

Neural Spline Flows — (NSF)

Instead of simple affine coupling layers, NSF replace each transformation with a monotonic rational-quadratic spline: the domain is split into K bins, each mapped by a rational quadratic function.

A MLP takes the features $\mathbf{x}$ and outputs the spline parameters so the entire distribution shape is determined by the input context.

→ more expressive, handles heavy-tails, fast inversion computation

Input features:

- Disaster type, magnitude, duration drives the actual intensity of the natural disaster
- Geographic features like slopes, altitude and distance from water basins drives the different complications the disaster can brings
- Different land coverities drives the exposure of the disaster
- Historical temperatures and precipitations drives ignition risk and drought

Output of the model

The output is a full predictive distribution $P(\mathbf{y}|\mathbf{x})$ and this enables:

- uncertainty quantification,
- risk-level prediction (e.g. 95th percentile damages),
- Monte Carlo sampling for downstream risk analysis.

02

Data Exploration & Feature Engineering

Exploratory analysis of the datasets: disasters and conditioning features.

Distribution of events, target variable characteristics, correlation analysis, and feature preprocessing steps.

Dataset: Disaster events

Source

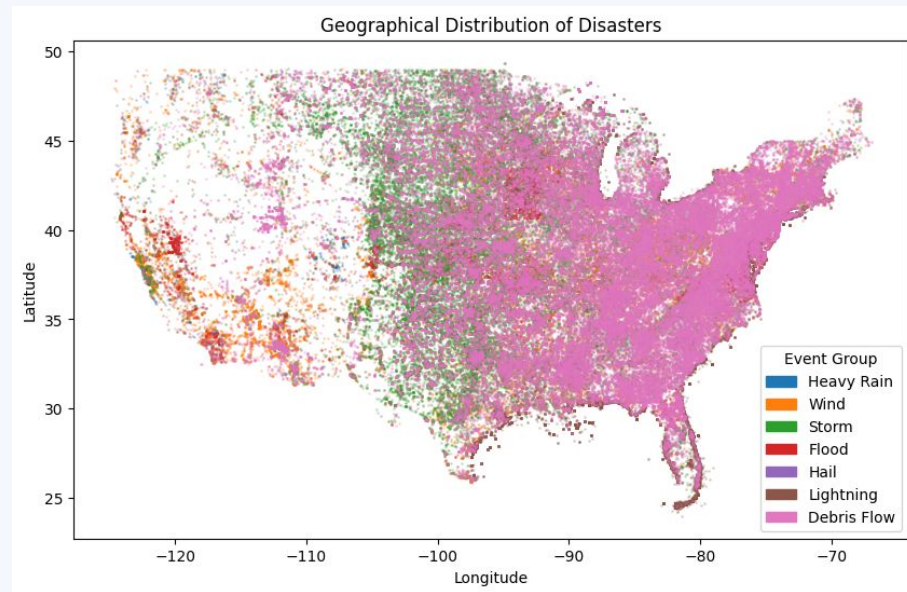
NCEI Storm Events Database
Years: 2016-2022 · N events: ~400 000

Preprocessing steps

- Remove all feats except: time feats, coordinates, targets, magnitude, type of event
- aggregate targets into CASUALTIES and DAMAGES
- aggregate event type into groups
- refine time features into `TIME_DAY_SIN`, `TIME_DAY_COS`, `TIME_YEAR_NORM`

Target Variables

Economic damages (\$) · Number of casualties (deaths + injuries)



Data Exploration

Target Distribution

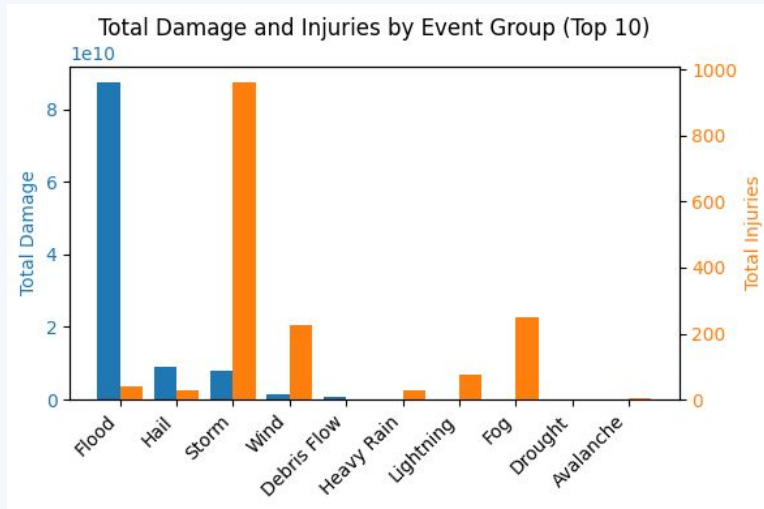
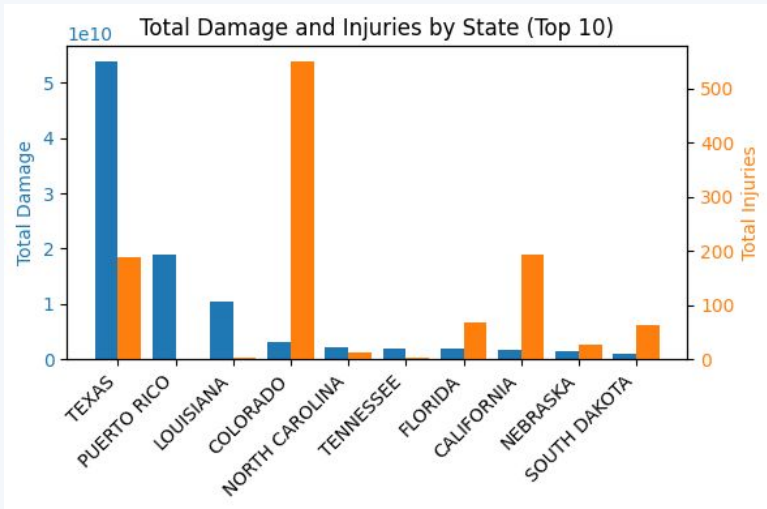
Highly skewed, heavy-tailed distributions.
Extreme events represent >80% of total damages.

Feature Engineering

Time feats with sin, cos and duration
Spatial feats with lat and lon

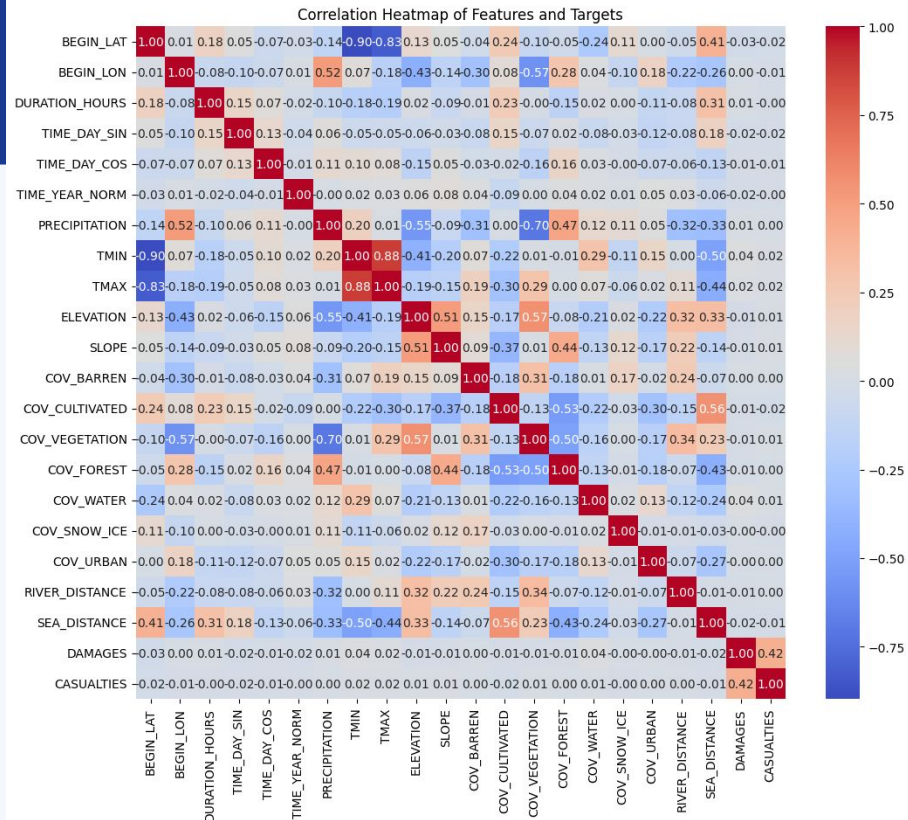
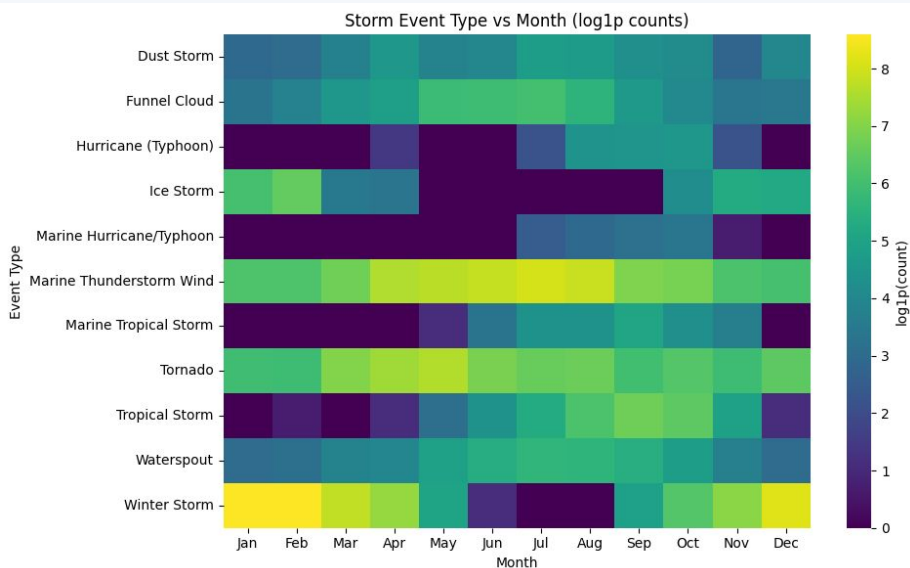
Data Normalization

Log1p on targets
Standardization on all feats



Data Exploration

Different disaster types greatly differs in terms of timing, locations, and effects. This is why we train a model per event group



Some feats correlates well by construction.
Generally the features subset provides good informations.

Dataset: climate

Source

PRISM & ERA-5 Land datasets

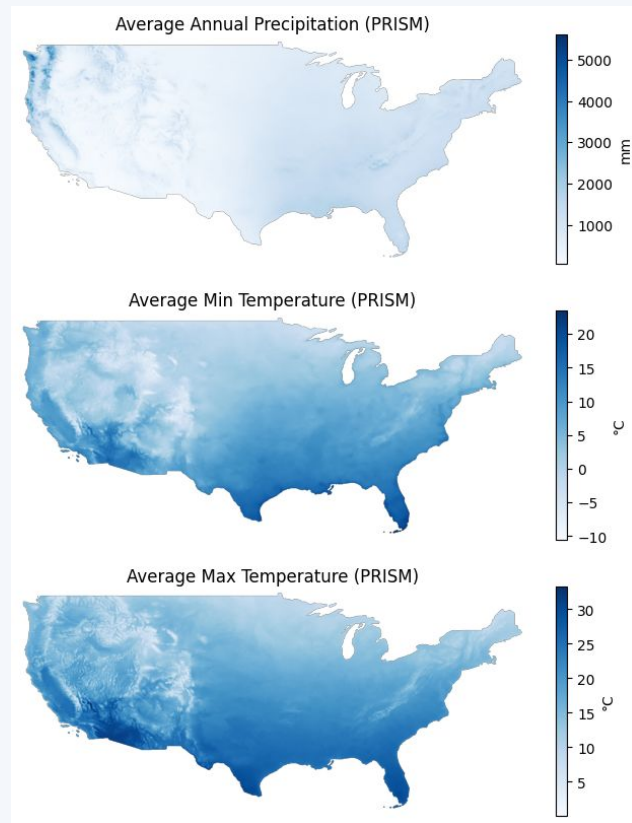
Aggregation

Annual averages

16 points inverse distance weighted interpolation

Output features

PRECIPITATION (mm), TMIN (°C), TMAX (°C)



Dataset: land cover

Source

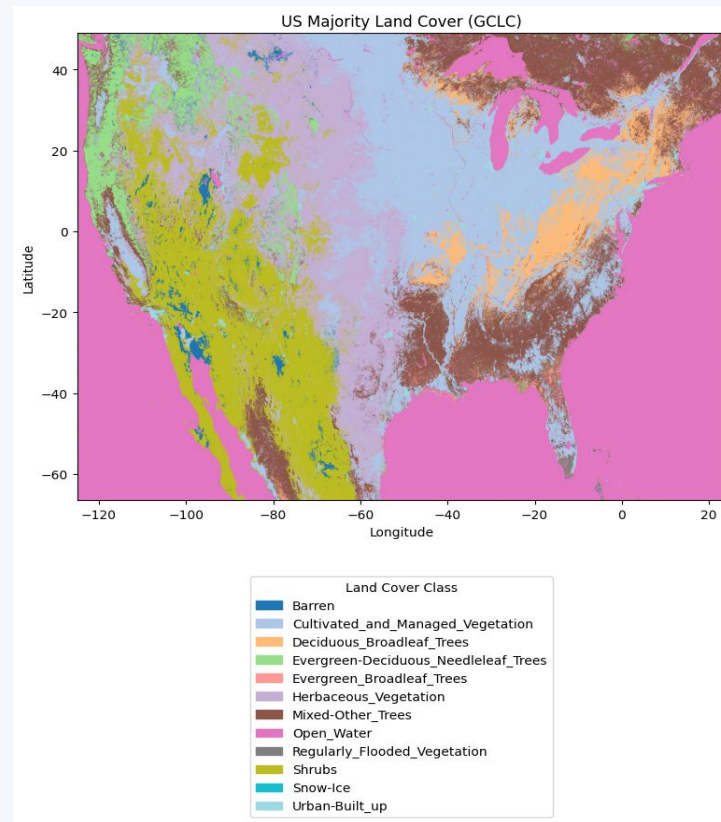
GCLC datasets (one per terrain type)

Computation

aggregation of all terrain types into land cover percentage

Output features

COV_BARREN, COV_CULTIVATED, COV_VEGETATION,
COV_FOREST, COV_WATER, COV_SNOW_ICE,
COV_URBAN



Dataset: morphology

Source

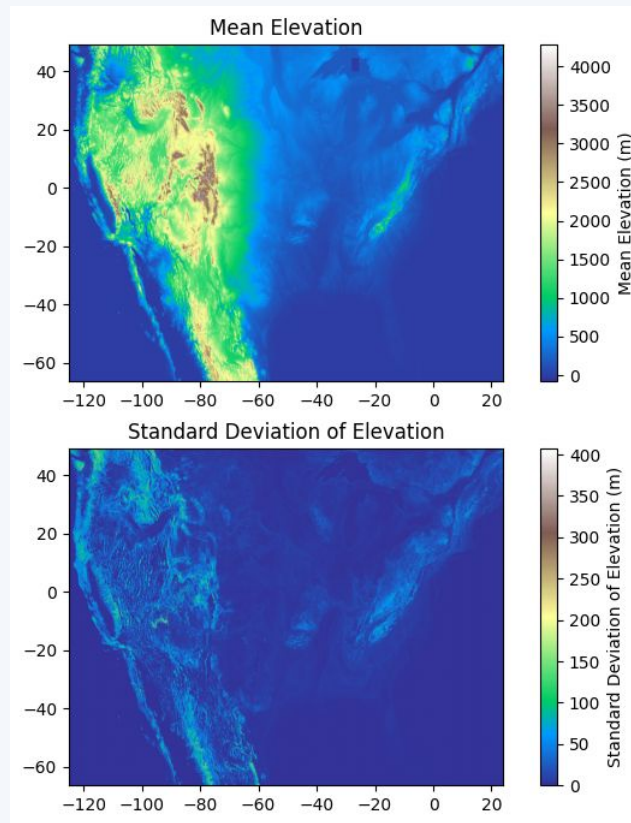
GMTED datasets

Computation

Closest point interpolation

Output features

ELEVATION (m), SLOPE (m)



Dataset: coastline

Source

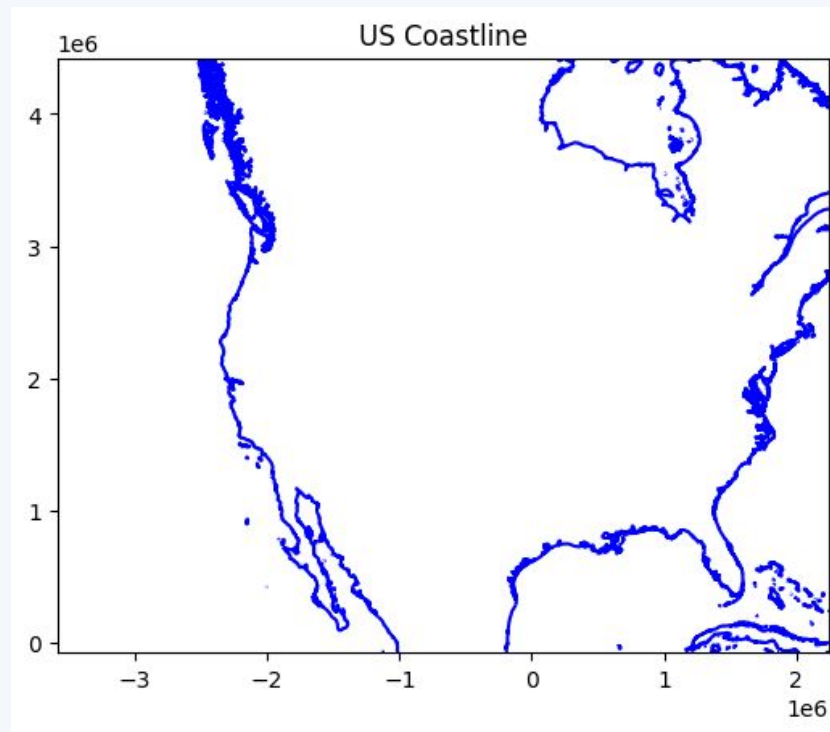
GSHHG dataset

Computation

Distance to closest coast point

Output feature

SEA_DISTANCE (km)



Dataset: hydrography

Source

GSW dataset

Computation

Distance to closest river point

Output feature

RIVER_DISTANCE (km)



03

Model & Training Insights



Deep dive into the specifications of our model and the training recipes we used.

Model Architecture

Parameterization via Rational-Quadratic Splines

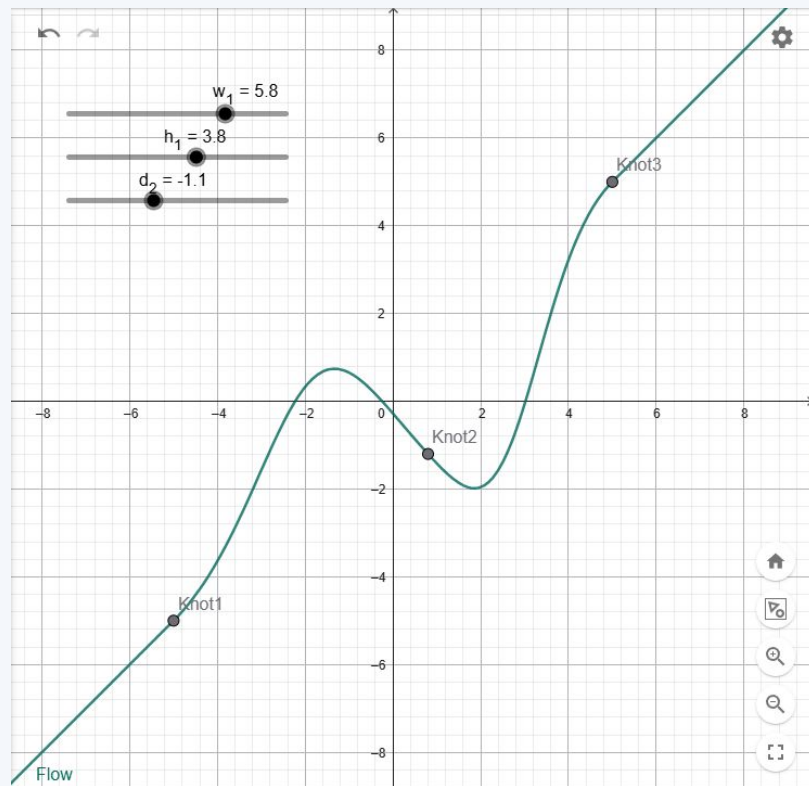
NSFs use monotonic rational-quadratic splines to parameterize these transformations. They are invertible and bounded, with linear tails for inputs falling outside their range, set as $[-5, 5]$ by default.

Conditioner Architecture and Parameter Generation

The needed transformation parameters are output by a 3 layer MLP with the conditional parameters as input and $3K - 1$ unconstrained values per dimension to define K bins within a specified bounding box.

Geometric Constraint Enforcement

These raw outputs are transformed through specific activation functions to satisfy the strict geometric and monotonic constraints of the spline: Widths, Heights and Knot Derivatives



Model Architecture

Architecture

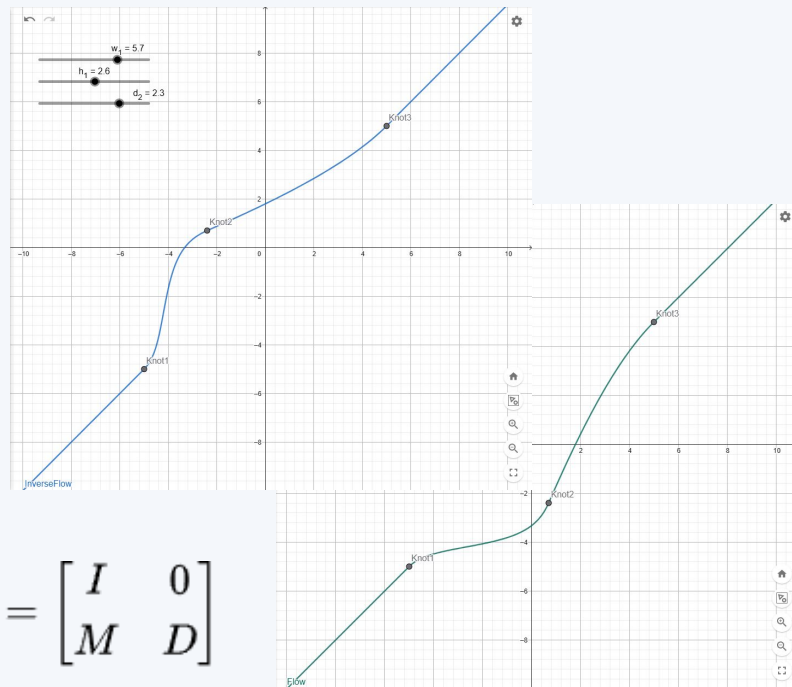
Coupling layers, Neural Spline Flows, 1.3 M parameters
Conditioner network: 3-layer MLP (256→256→256→47)
Flow depth: 8 bijective transformations, each with 16 bins
Target latent distribution: Bidimensional standard Gaussian $N(0, I)$

Input Partitioning

The input vector is partitioned into two disjoint subsets. At each layer, one subset remains static while the other undergoes the spline transformation. This active transformation is conditioned on both the unmodified static subset and the external conditioning variables.

Tractable Determinant and Layer Alternation

Since the active transformation depends only on the unmodified partition, the layer's Jacobian matrix is strictly lower-triangular. Consequently, its determinant reduces efficiently to the product of its diagonal terms, enabling rapid log-likelihood computation.



$$J = \begin{bmatrix} I & 0 \\ M & D \end{bmatrix}$$

$$\det(J) = \det(I) \times \det(D)$$

Forcing the Gaussian Prior: The Role of NLL Loss

The Objective

NSFs are optimized via Maximum Likelihood Estimation, minimizing the Negative Log-Likelihood (NLL). This is equivalent to minimizing the Forward KL Divergence between the true data distribution and the model.

The Gaussian Prior

The vase distribution $p(z)$ is explicitly hardcoded as a standard multivariate Gaussian, $N(0, I)$. Its log-density is strictly proportional to:

$$-\frac{1}{2} \|z\|^2$$

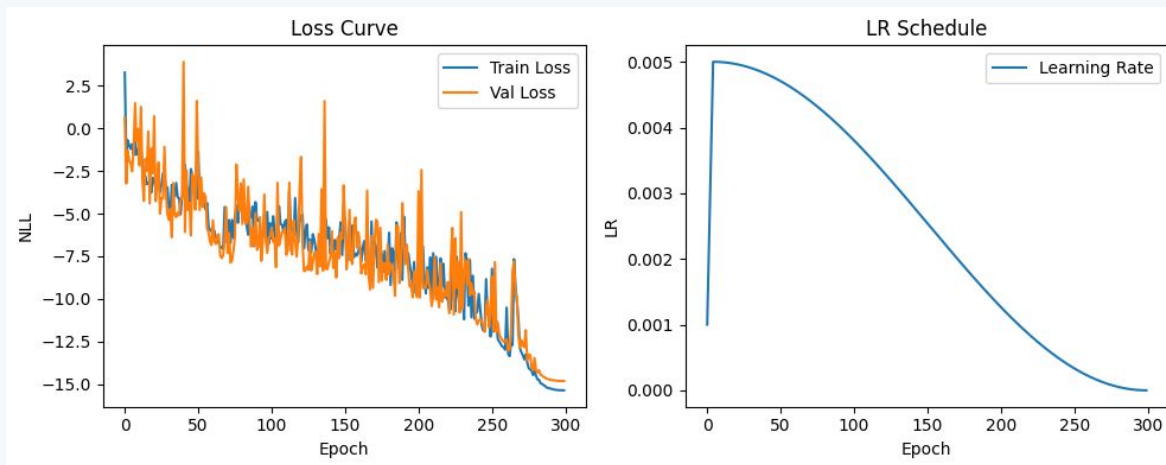
Change of Variables Formula

$$\log p_{\theta}(x) = \log p_z(f_{\theta}(x)) + \log |\det J_{f_{\theta}}(x)|$$

Opposing Optimization Pressures

- Pressure 1: The Pull to the Center:
to minimize the overall loss, the network must maximize the prior likelihood $\log p_z(f_{\theta}(x))$. Because of the Gaussian definition, this applies a strict quadratic penalty to any latent vector z placed far from the origin, actively forcing mapped data points into a normal distribution curve.
- Pressure 2: Volume Expansion (Anti-Collapse):
The network cannot simply map all x to $z=0$ to perfectly satisfy Pressure 1. It must simultaneously maximize the log-determinant of the Jacobian ($\log |\det J_{f_{\theta}}(x)|$). This term penalizes extreme spatial contraction, forcing the points to spread out and maintain the unit variance dedicated by the prior.

Training Details



Training Setup

Optimizer: Adam · LR: $1e-3$ (cosine decay)
Batch size: 512 · Epochs: 300
Loss: Negative log-likelihood (NLL)
Early stopping on validation NLL

Data Split

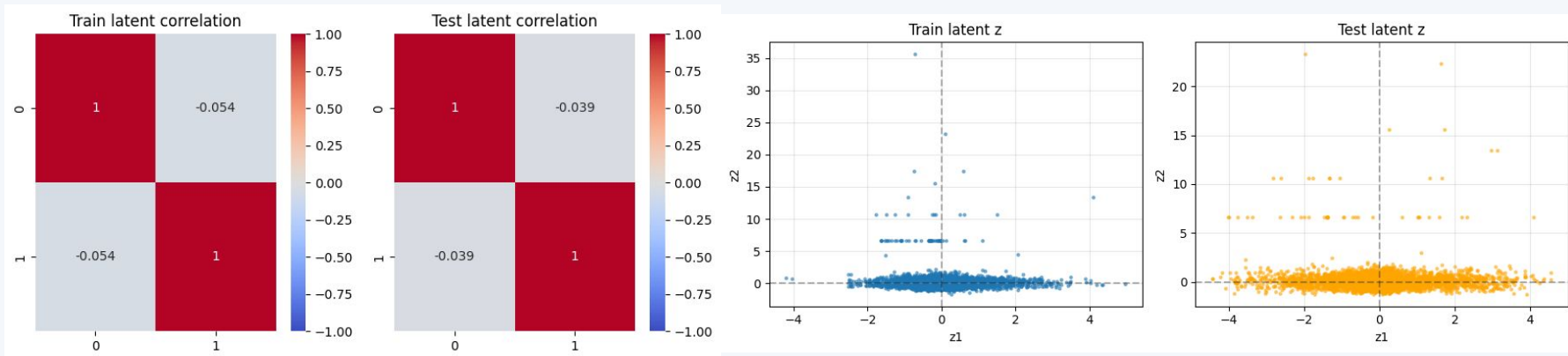
Train 70%
Validation 15%
Test 15%

04

Results Analysis

Quantitative evaluation on the held-out test set.

Gaussian assumption test - Floods

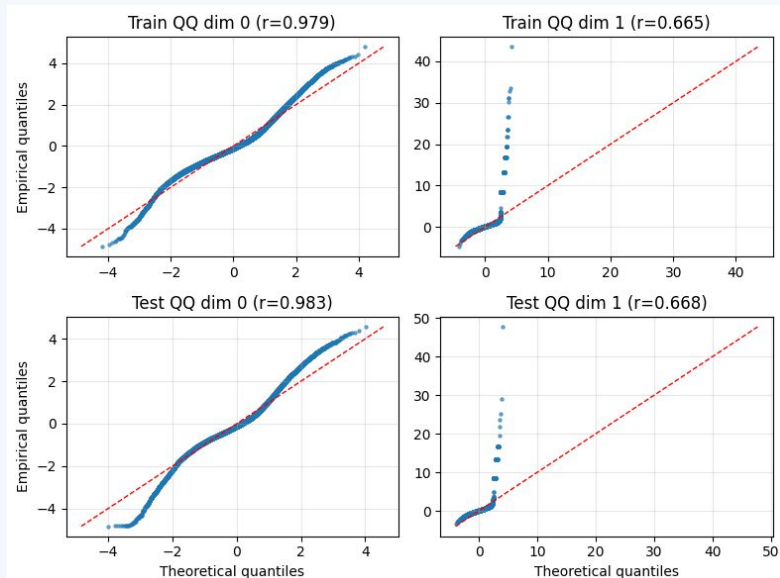
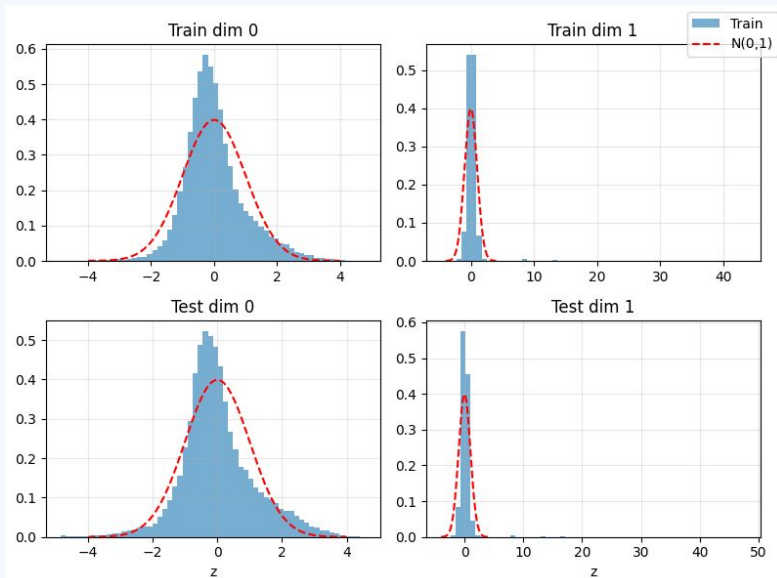


(Flood model)

Training and test set points, mapped into the latent space by the trained model.

Output is correctly Gaussian-like but the discrete and zero-skewed nature of the casualty variable collapses the probability distribution along its direction in a way that annuls the Jacobian.

Gaussian assumption test - Floods



(Flood models)

The data marginal distributions are also approximately gaussian, with some heavy tails behaviour for damages and extremely anomalous distant tails for the casualties.

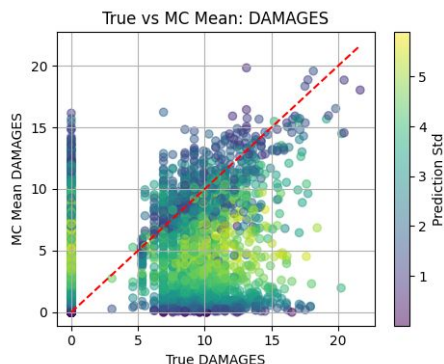
Sampling Procedures - Floods

Monte-Carlo Sampling

Generates samples by drawing base noise (from a normal distribution) and applying the conditioned inverse flow.

Spline inversion is analytical, solving quadratic roots per bin without numerical approximation. Yields exact, unbiased sample from $p(x|c)$ and explores full distributional support.

MSE for DAMAGES: 15.63

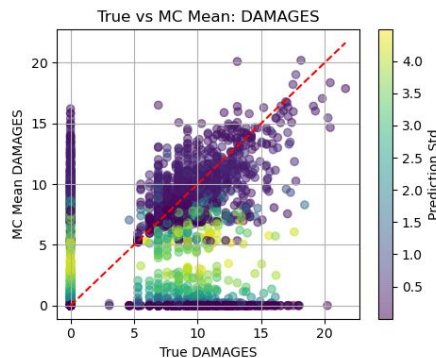


Deterministic Radical Lattice

Evaluates exact density $\log q_\phi(x | c)$ or maps z-space across a strictly regular, multidimensional grid.

Guarantees deterministic, noise-free density coverage for exact numerical integration or confidence interval mapping.

MSE for DAMAGES: 18.10



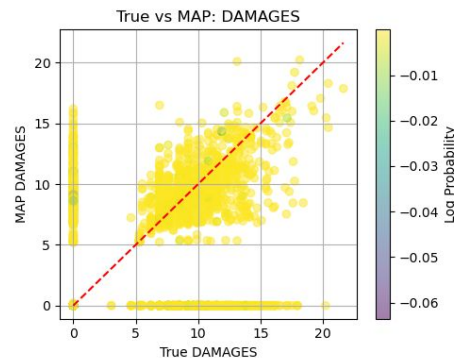
Maximum Likelihood

Solves for the exact mode:

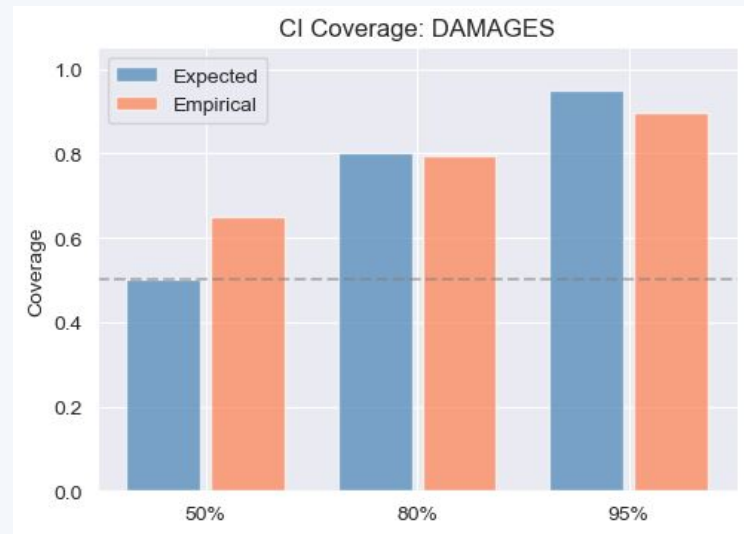
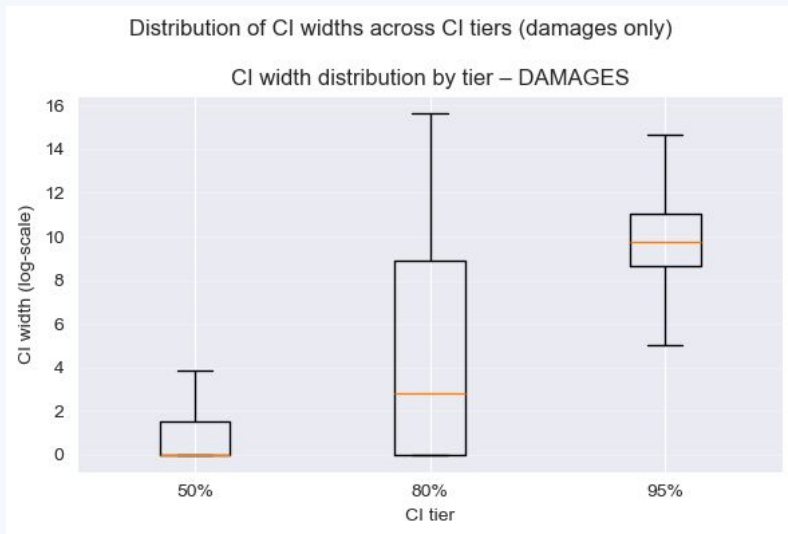
$$x^* = \arg \max \log q_\phi(x | c)$$

Optimizes x directly via exact forward density backpropagation from multiple starting points. Fails to capture uncertainty and is strictly local, heavily prone to trapping in local maxima within highly multimodal NSF distributions

MSE for DAMAGES: 18.71



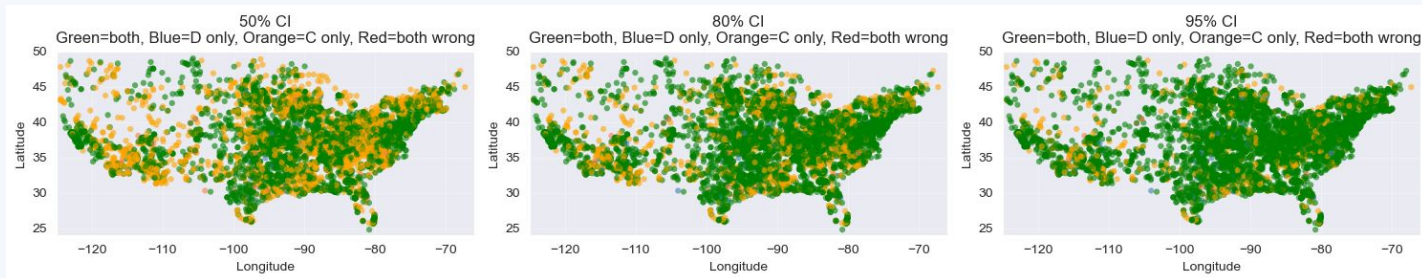
Coverage Interval Results - Floods



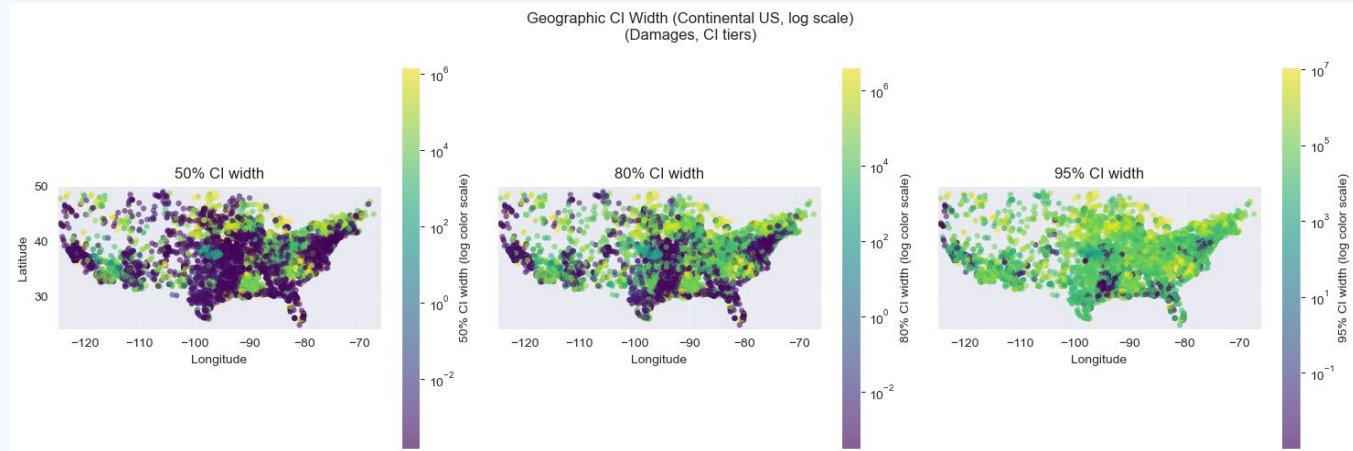
The zero-skew of the distribution is readily apparent when enlarging the confidence interval range, both in confidence interval width and predictive power.

In the case of casualties, the model entirely collapses to exclusively predicting the null label.

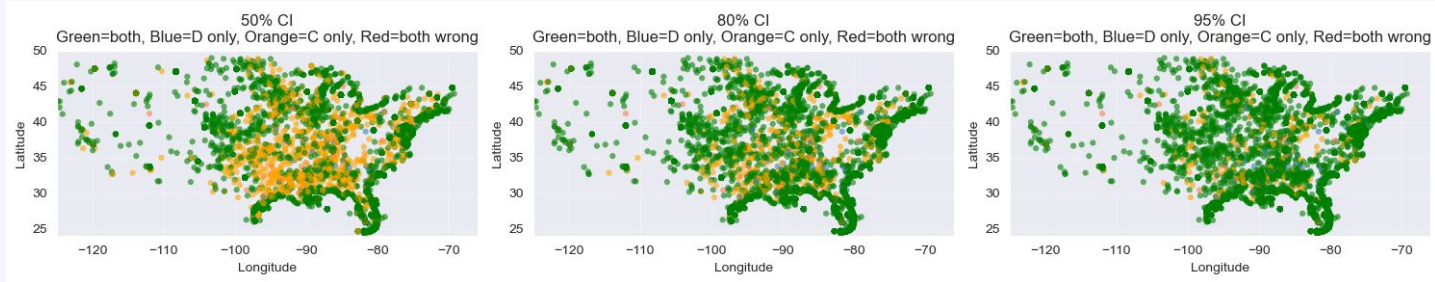
Locality of predictions - Floods



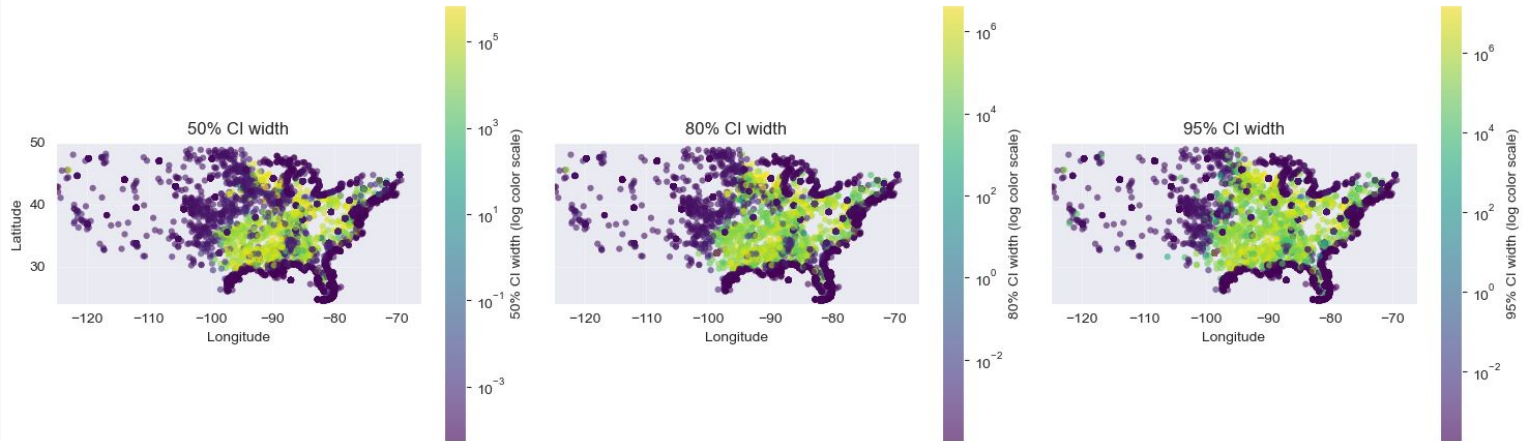
Mississippi River Basin



Locality of predictions - Storms



Geographic CI Width (Continental US, log scale)
(Damages, CI tiers)



05

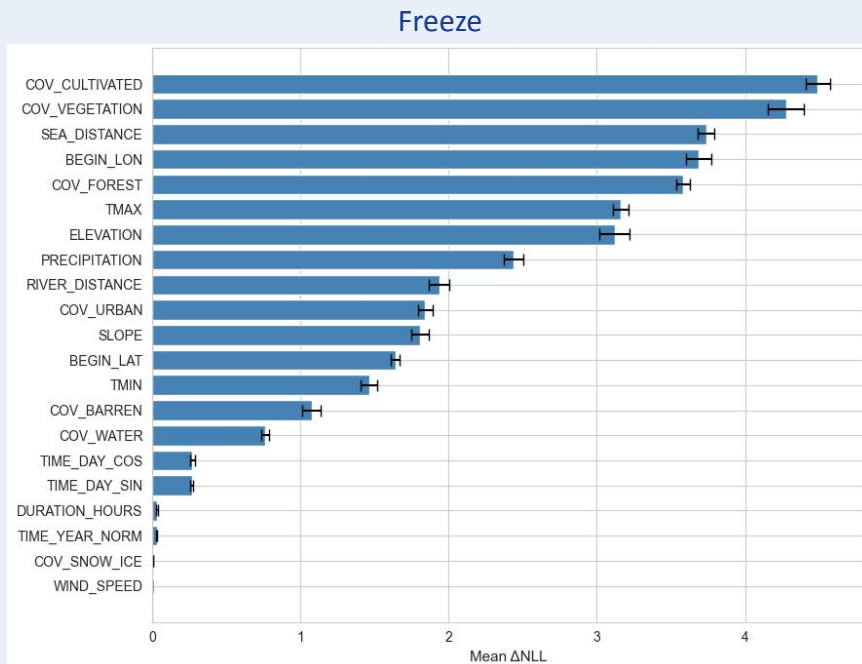
Feature Importance Analysis

Which features drive damage and casualty predictions?
Which metrics can we use to better understand features contributions?

Permutation Feature Importance

Remove the contribution in the context of a single feature making it as pure noise, analyse then the difference in the NLL.

To break all the dependencies of such feature (with the targets and with the other features) we permute all its values from the test set several time and we average the observed difference in the NLL.



Top Predictors — Damages

Different land coverage · sea distance
temperatures · wind speed (only in some models)

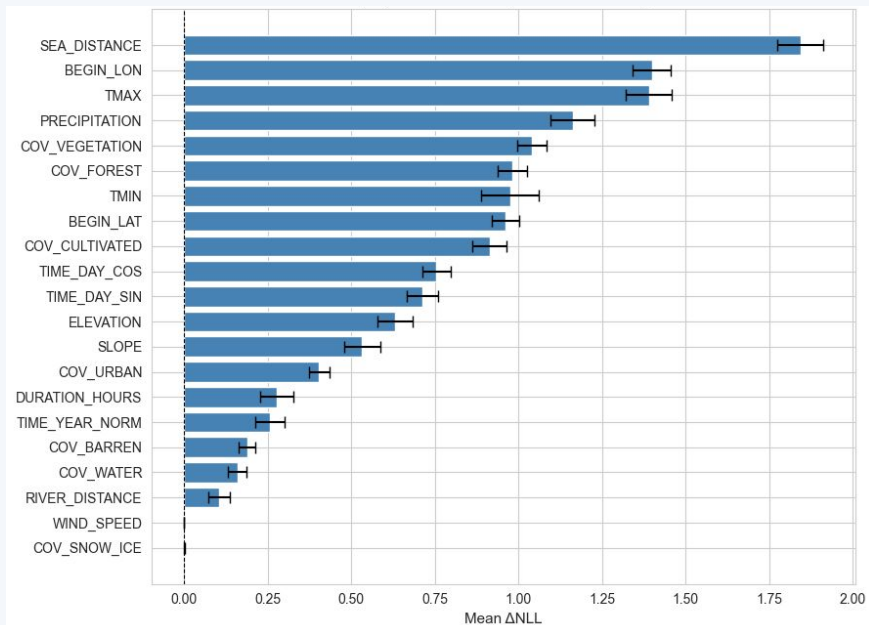
Note

Feature importance varies significantly by disaster type, and it emerges different reasonable patterns:

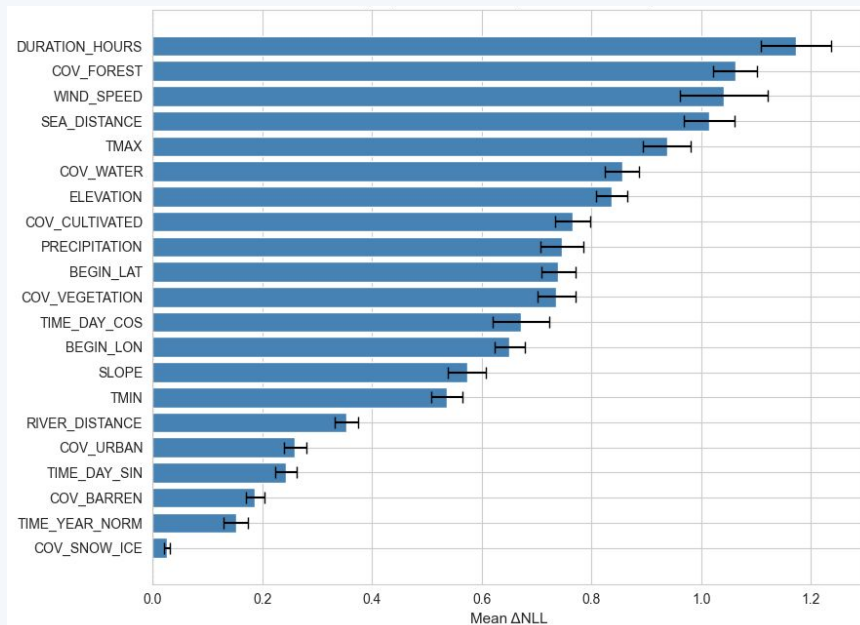
- sea distance and precipitation strongly influence flood model
- wind speed, duration strongly influence the storm model
- coverage cultivated is the top predictors for damages in the freeze model

Permutation Feature Importance

Flood



Storm



Jensen-Shannon Divergence

Why this metric?

Our model produce a predicted distribution of the target which we analyse with Monte Carlo sampling. PFI measure the change in the NLL, which can be misleading with heavy-tail distribution.

Output

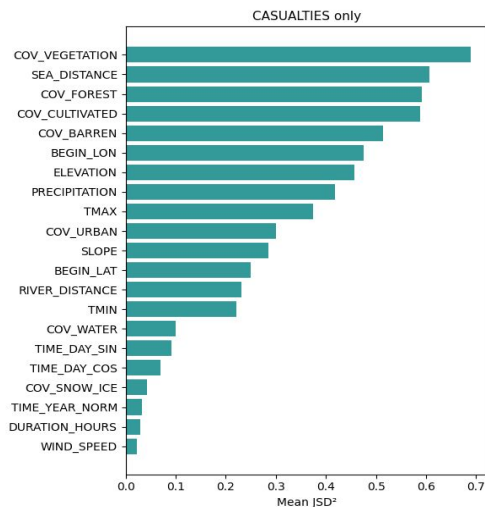
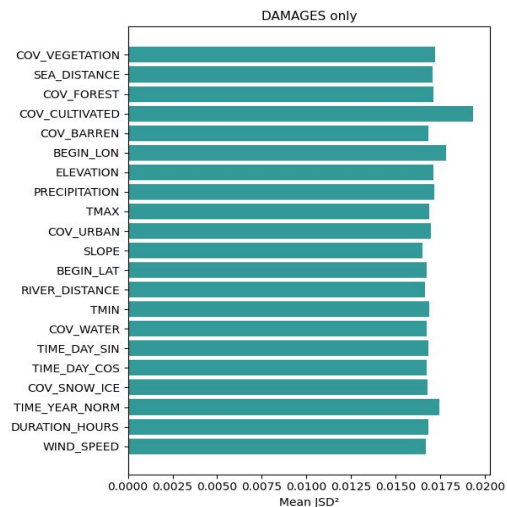
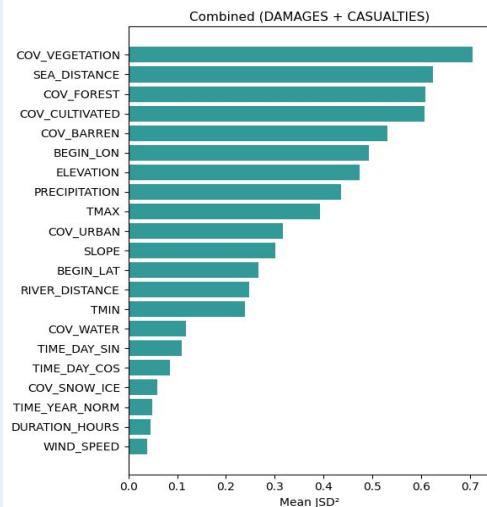
It measure the distributional impact of a feature over the predicted target distributions.

Computation

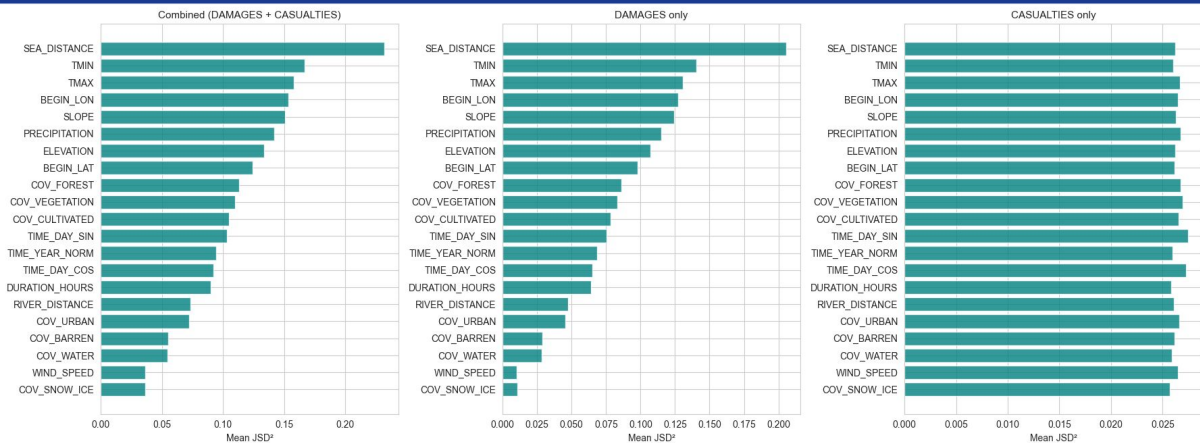
Given two distribution p and q :

$$JSD(p||q) = \frac{1}{2}KL(p||m) + \frac{1}{2}KL(q||m),$$

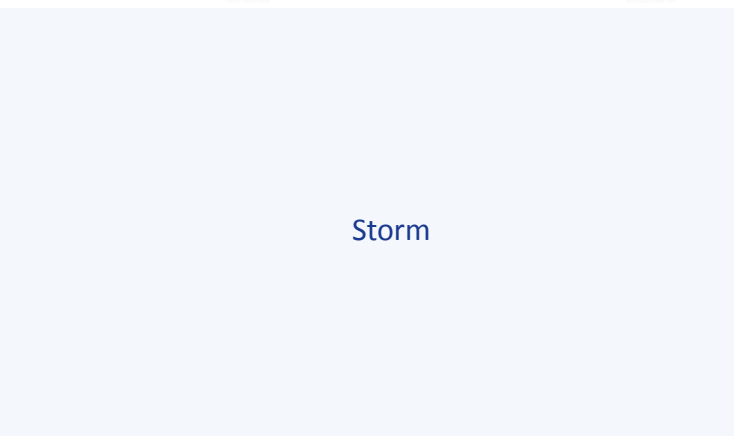
$$\text{where: } m = \frac{p+q}{2}$$



Jensen-Shannon Divergence



Flood



Storm

Individual Conditional Expectation — ICE curve

Why this metric?

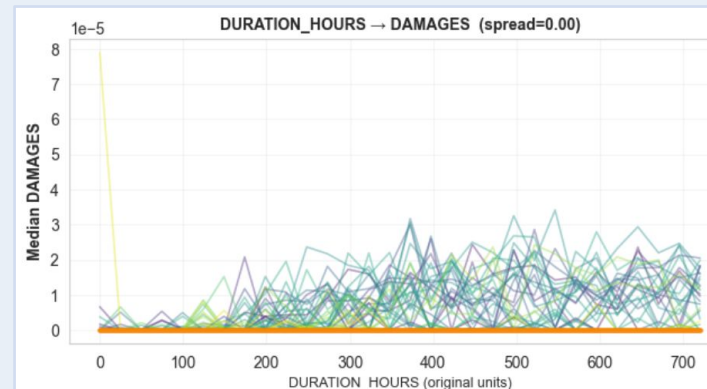
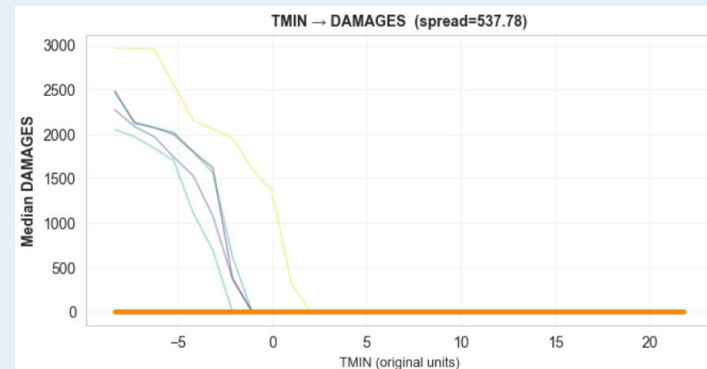
Unlike aggregated methods ICE curves reveal if the feature's effect is consistent across all predictions. It provide insights into how a model makes a prediction for a specific event or data point.

Computation

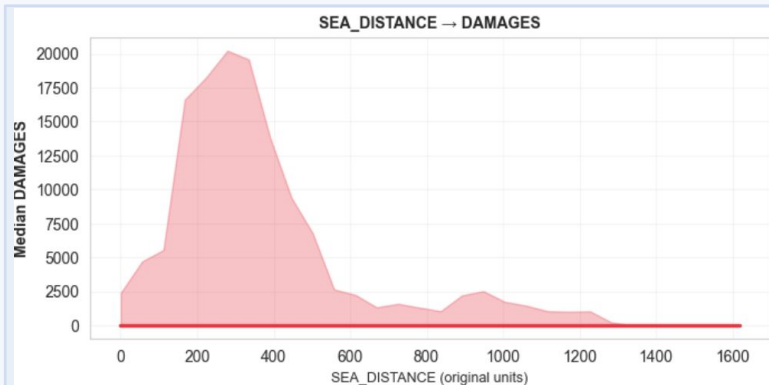
1. Select a single instance,
2. fix all features except the feature of interest,
3. vary the feature of interest over its entire range,
4. record the model's prediction for each varied value,
5. repeat this process for multiple instances to generate multiple curves.

Output

- Divergence → different instances respond differently to changes in the feature (heterogeneous effect)
- Change in the direction → the effect of the feature is not constant across its range.

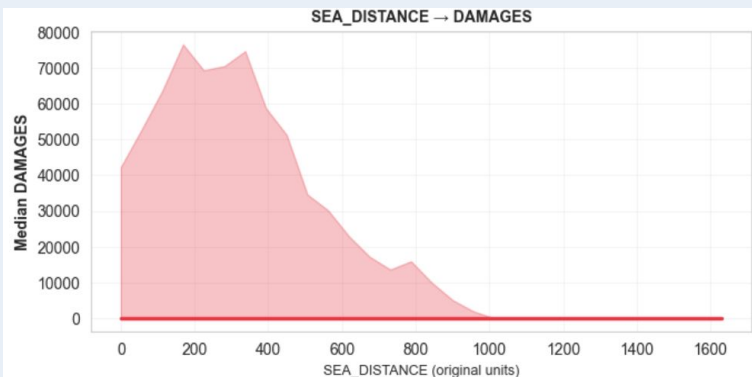


Partial Dependence Plot — PDP



Flood

Storm



Why this metric?

PDP reveals feature's average marginal effect. Output shows the directional impact of a feature on the prediction.

Computation

Averages the median of the model predictions over many samples that differs only in one feature across a defined range.

Output

- Flat line: feature has negligible marginal effect
- Rising/falling trend: directional impact on severity
- Wide IQR band: heterogeneous effect across events (feature interacts with other covariates)

06

Training Improvements & Side Analysis

What could be improved? Weighted training for extreme events, and a comparative study: one global model vs. one specialized model per disaster type.

Training improvements: what had an impact

Model

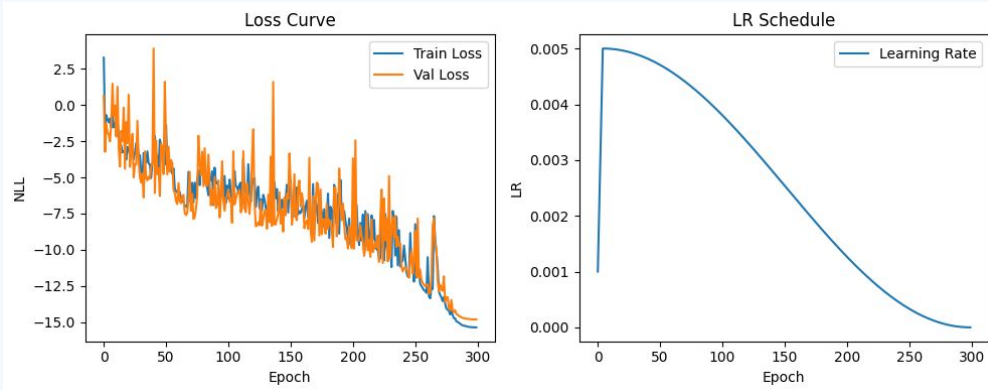
- Increasing the n. of layers ✓
- Increasing the n. of parameters ✗
- Increasing the n. of transforms ✗
- Increasing the n. of bins ✓

Training

- Mixed precision ✗
- Warmup + cosine scheduling ✓
- Gradient clipping ✓

Data

- Target (casualties) dequantization ✗
- Dataset balancing through undersampling of zero impact events ✓ (minor improvement)



An alternative loss

Right now

Our model treats all events equally.

Why is it a problem?

Most of the events have 0 value targets (our dataset is unbalanced)

A good enough model could just predict always 0

Why not using a cost based training?

The model would collapse and lose its distribution estimation capability

PAL loss

Idea: assign higher loss weight to events in the top 5-10% of damages/casualties.

Expected benefit: better calibration and coverage on rare, catastrophic events.

Updated Loss: $L = w \times NLL$, $w = 1 + \lambda_1 e_1 + \lambda_2 e_2$

The calibration issue: this loss requires many more parameters to tune specifically for every single model.

What is it that we want to minimize? NLL, damages, casualties?

Each one of them requires a different choice of params.

Side analysis

One Model per Disaster Type vs. One Global Model

One Model per Disaster Type

This was the initial idea, as the we thought that each disaster type would benefit from a specific estimator.

- ✓ Specialized to each distribution
- ✓ Needs less generalization
- ✗ Requires more data per type
- ✗ Requires separate training

The analysis of the results, though, suggested that many of the estimators mostly relied on the same subset of features.

This led us to think a the single model approach was feasible

VS

Single Global Model

Which data is to use? The disaster types are not well balanced.

⇒ for the best results balance where possible.

How many data points? Adding up all the data would result in a very large set.

⇒ Sample 200000 points

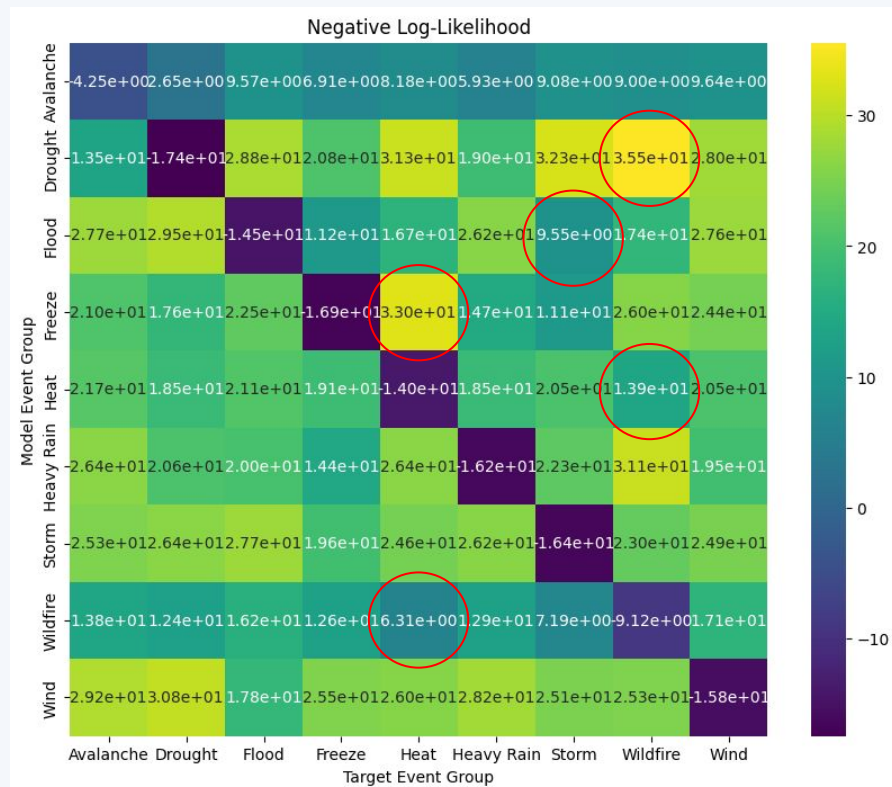
What architecture?

⇒ the same for multiple models, for comparison

- ✓ Shared representation across types
- ✓ Less data needed per type
- ✗ No distinction between types

Result: the achieved NLL is very similar, but the single model learns a worse representation of the data

Cross inference: an interesting find



What is cross inference?

What is here referred as cross inference is evaluating a model trained on a dataset A with a dataset B, A and B of different event groups

Meaning

A model able to perform well on a different group highlight a similarity in the underlying patterns of the data

Result

Unsurprisingly freeze is a bad predictor for heat and flood is good for storm
Surprisingly drought is a bad predictor for wildfire.
Heat is a good wildfire predictor, and vice versa.

07

Conclusion

Summary of findings, contributions, and open research questions.
What we learned from applying probabilistic modeling to natural disaster impact estimation.

Conclusion



Conditional Flow Model — Better captures the heavy-tailed, learns physically meaningful conditional distributions



The casualty collapse — it is a discrete, zero-inflated count variable. The continuous normalizing flow, which assumes a differentiable density, cannot natively represent a point mass at zero.



Probabilistic metrics — NLL, coverage intervals, and JSD — The coverage results show the model is well-calibrated for damages but collapses for casualties. This evaluation framework is itself a contribution: it establishes the right benchmark for future work on probabilistic disaster impact prediction.



Future Work — Data filtering, improved model, richer spatial representation, two component casualty model

Thank you

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Questions?